

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

UNIT AND MEASUREMENTS

LECTURE-9

PHYSICS

WAJAHID SIR (Gabbar Sir)



Topics To Covered

1 Revision

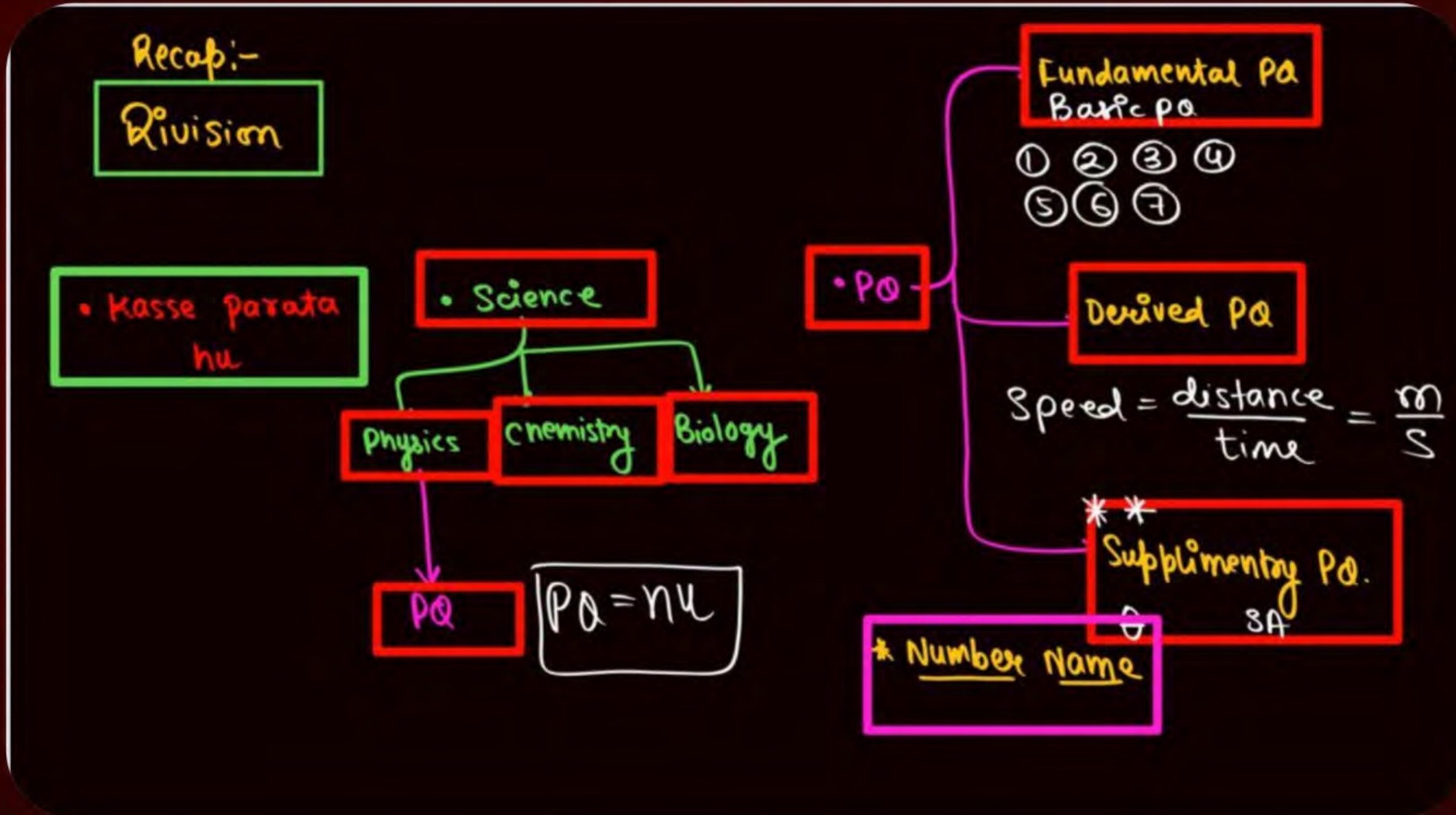
2 PYQs

3 ..

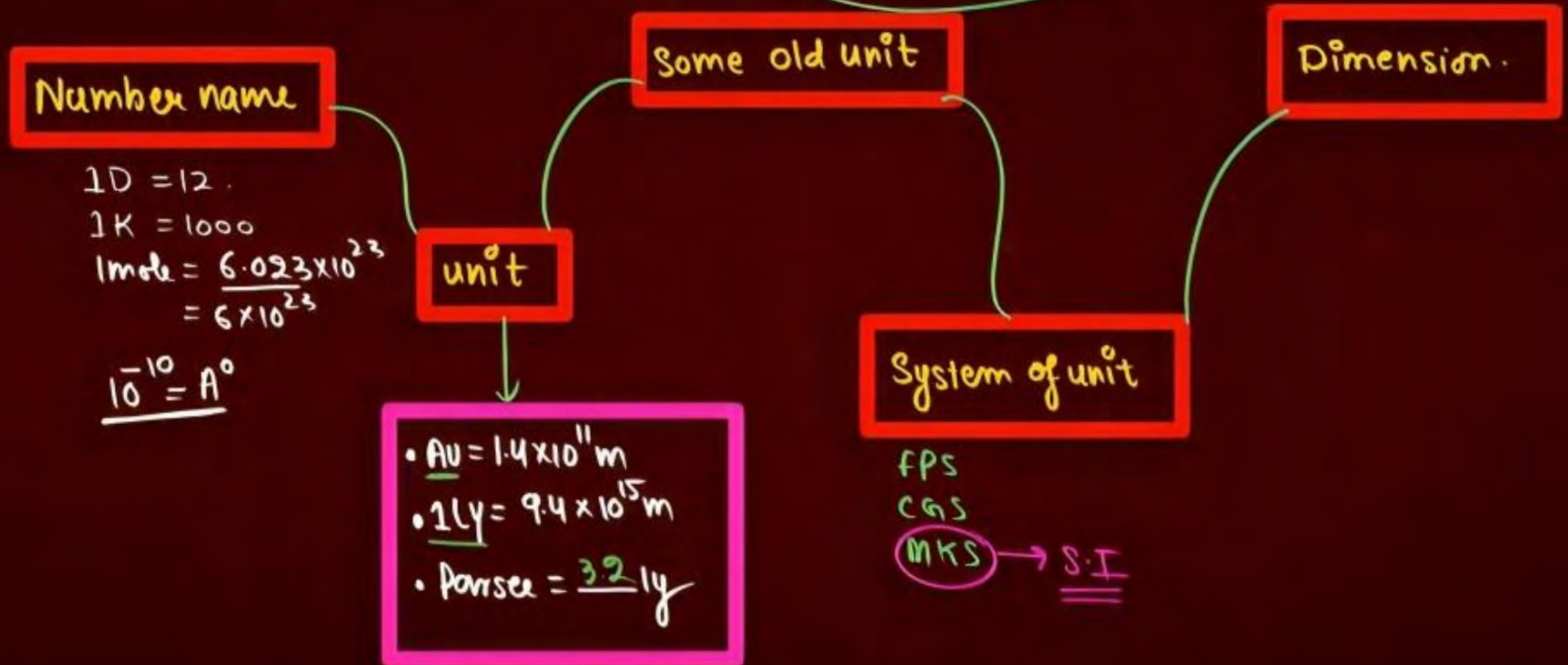
4 ..

“Insaan nahi bolta,
uske **din bolte** hain lala
Or jab din bolte hain
To behre bhi sunte hai”

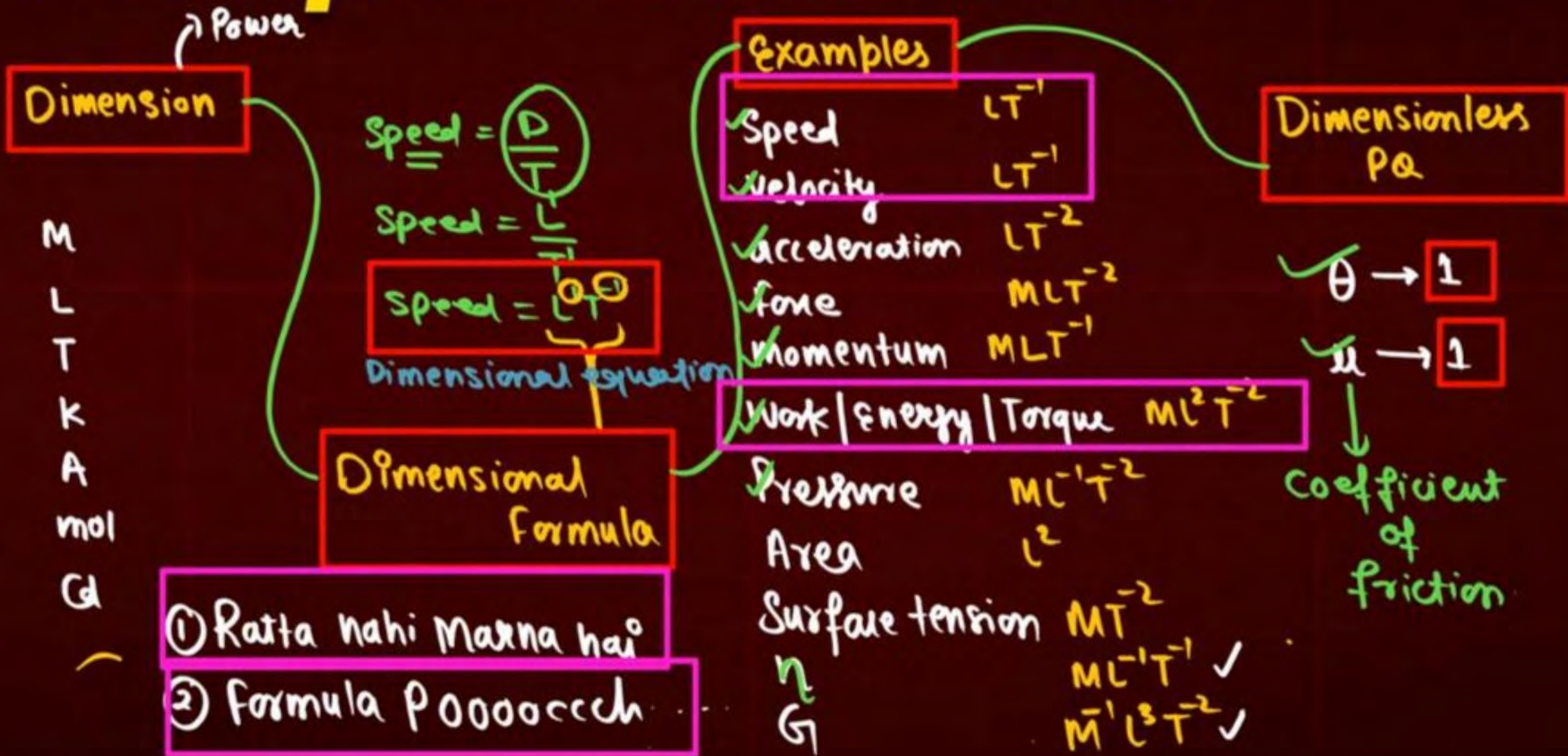
Recap From 1st Lec.



Recap From 2. Lec.



Recap From 3rd Lec.



Recap From 4th Lec.

Application
of
Dimension

a ✓

b ✓

c

d

a To find out the
Dimensional
Formula

$$\eta \rightarrow \text{ML}^{-1}\text{T}^{-1}$$

$$G \rightarrow \text{ML}^3\text{T}^{-2}$$

b To check the
Correctness of
Physics Formula

Law of
Homogeneity

$$x + x = 2x$$

$$2x - x = x$$

① $v = u + at$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$\text{LT}^{-1} \quad \text{LT}^{-1} \quad \text{LT}^{-1}$$

② $s = ut + \frac{1}{2}at^2$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$\text{L} \quad \text{LT}^{-1} \quad \text{LT}^{-1}$$

③ $v^2 = u^2 + 2as$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$\text{L}^2\text{T}^{-2} \quad \text{L}^2\text{T}^{-2} \quad \text{LT}^{-2}$$

④ $F = \frac{mv^2}{r}$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$\text{MLT}^{-2} \quad \text{MLT}^{-2} \quad \text{MLT}^{-2}$$

Recap From 5th Lec.

Application of
Dimension

© To derive the
physics formula

Time period of Pendulum

• let
 $T \rightarrow m^a l^b g^c$
 $M^0 L^0 T^1 \rightarrow M^a L^b (LT^{-2})^c$
 $a=0 \quad b=\frac{1}{2} \quad c=-\frac{1}{2}$



$2T = 2\pi \sqrt{l/g}$
 $T = 2\pi \sqrt{l/g}$

Stroke law

• let
 $F \rightarrow \eta^a r^b v^c$
 $MLT^{-2} \rightarrow (ML^{-1}T^{-1})^a L^b (LT^{-1})^c$
 $a=1 \quad b=1 \quad c=1$

$F = 6\pi \eta r v$

Recap From 6th Lec.

④

conversion of
one unit system
into another
unit system

- * $1\text{m} = 100\text{cm}$
- * $1\text{kg} = 1000\text{gm}$
- * $1\text{N} = 10^5\text{dynes}$ (PVA)
- * $1\text{joule} = 10^7\text{erg}$ (PVA)

$$n_2 = n_1 \left[\frac{M_1}{M_2} \right]^a \left[\frac{L_1}{L_2} \right]^b \left[\frac{T_1}{T_2} \right]^c$$

* Limitation of Dimension

- * we cannot derive the Dimensional const.
- * we cannot derive the formula having $(+/-)$
- * Cannot derive the formula with $\sin\theta / \cos\theta$
- * It is not unique

Recap From 7th Lec.

Rounding off

Rule ① 2.4829
 2.483

Rule ② 2.4562
 2.456

Rule ③ 5.234215 5.234225
 5.23422 5.23422

Significant Fig

Number

SF

24321

5

24000

2

34051

5

4.600

4

0.009

1

3×10^{-6}

1

number

∞ (infinite)

Recap From 8th Lec.

Scientific Notation

* Ek tarika hai to reduce the number

$$245620. = 2.4562 \times 10^5$$

$$0.00052 = 5.2 \times 10^{-4}$$

Gabbbar

Numerical Series

PYQs

QUESTIONS[WINTER SERIES-A 2025]

The number of significant figure in 2.64×10^{24} Kg is:

1

3 ✓

2

4

3

24

4

1

$$\underline{2.64} \times 10^{24}$$

$$SF = 3$$

QUESTIONS[WINTER SERIES-A 2025]

The correct relation between Light year and metre is:

1 1 Light year = 7.469×10^{15} m

2 1 Light year = 4.2 m

3 1 Light year = 9.467×10^{15} m

4 None of them

QUESTIONS[WINTER SERIES-A 2025]

Write the uses and limitations of the dimensional analysis.

#

uses

limitation

QUESTIONS[WINTER SERIES-B-2025]

The number of significant figure in 6.032 Nm^{-2} is:

6.032

1

6

2

3

3

2

4

4

QUESTIONS[WINTER SERIES-B-2025]

Which one is the correct relationship between Angstrom and Astronomical unit?

1 $1 \text{ A}^\circ = 10^{-10} \text{ m}$

2 $1 \text{ AU} = 1.3 \times 10^{11} \text{ m}$

3 $1 \text{ A} = 1.496 \times 10^{11} \text{ m}$

4 None of them

$$1 \text{ A}^\circ = 10^{-10} \text{ m}$$

$$1 \text{ AU} = 1.4 \times 10^{11} \text{ m}$$

$$= 1.496 \times 10^{11} \text{ m}$$

QUESTIONS[WINTER SERIES-C-2025]

The number of significant figure in 0.0006032 is:

0.0006032

1 6

2 3

3 2

4 4

QUESTIONS [SUMMER SERIES-A- 2025]

Which of the following is not the unit of surface tension?

1 N/m ✓

2 J/m² ✓ $ST = \frac{F}{l} = \frac{MLT^{-2}}{L} = \underline{MT^{-2}} \rightarrow \frac{J}{m^2} = \frac{ML^2T^{-2}}{L^2} = MT^{-2}$

3 Kg/s² ✓

4 N/m²

$N/m^2 = \frac{MLT^{-2}}{L^2} = MT^{-2}L^{-1} = ML^{-1}T^{-2}$

$\frac{kg}{s^2} = \frac{M}{T^2} = MT^{-2}$

QUESTIONS [SUMMER SERIES-A- 2025]

-3 mark

Find the dimensions of Planck's constant. If its value in CGS system is 6.62×10^{-27} erg-sec. What will be its value on MKS system?

S.I

Sol $E = h\nu$ $\nu = \frac{1}{T}$

$$E = h \frac{1}{T}$$

$$\underline{ET} = h$$

$$ML^2T^{-2}T^1 = h$$

$$ML^2T^{-2+1} = h$$

$$ML^2T^{-1} = h$$

→ angular momentum

$$h = 6.62 \times 10^{-27} \frac{\text{erg}}{\text{sec}} (\text{C.G.S})$$

$$h = 6.62 \times 10^{-27} \times \underline{10^{-7}} \text{ Joule sec (S.I)}$$

$$h = 6.62 \times 10^{-27-7} \text{ Joule sec}$$

$$h = 6.62 \times 10^{-34} \text{ Joule sec}$$

$$1 \text{ Joule} = 10^7 \text{ erg}$$

$$1 \text{ Joule} = 10^7 \times \text{erg}$$

$$\frac{1}{10^7} \text{ Joule} = 1 \text{ erg}$$

$$10^{-7} \text{ Joule} = 1 \text{ erg}$$

QUESTIONS [SUMMER SERIES-B- 2025]

(New)



Which two of the following four physical parameters have same dimensions?

(i) Energy density

~~(ii) Dielectric~~

~~(ii) Refractive index~~

(ii) Young's modulus

1 ~~(ii) and (iv)~~

2 ~~(iii) and (iv)~~

3 (i) and (iii)

4 (i) and (iv)

$$\rightarrow \text{Energy density} = \frac{E}{V} = \frac{ML^2T^{-2}}{L^3} = \underline{ML^{-1}T^{-2}}$$

$$\rightarrow \text{refractive index} = \frac{c}{v} = \frac{\cancel{LT^{-1}}}{\cancel{LT^{-1}}} = 1$$

$$\rightarrow \text{Dielectric const} = \frac{\cancel{\cancel{L}}}{\cancel{\cancel{L}}} = 1$$

$$\rightarrow \text{Young modulus} = \frac{\text{Stress}}{\text{Strain}} = \frac{F/A}{1}$$

$$= \frac{F}{A} = \frac{m \times T^{-2}}{L^2}$$

$$= \underline{\underline{M L^{-1} T^{-2}}}$$

QUESTIONS [SUMMER SERIES-B- 2025]

The number of significant figures in 2.64×10^{27} kg is:

$\underline{2.64} \times 10^{27}$
 $SF = 3$

1

1

2

3

3

4

4

5

QUESTIONS [SUMMER SERIES-C- 2025]

The physical quantities not having same dimensions are:

$$P = mv = MLT^{-1}$$

$$h = ML^2T^{-1}$$

1 Momentum and Planck's constant

2 Speed and $(\mu_0 \epsilon_0)^{-1/2}$ Speed

3 Speed and $\sqrt{P/\rho}$ Speed of sound

4 Surface tension and spring constant

Surface tension

$$S = \frac{F}{l}$$

$$ST = \frac{MKT^{-2}}{L}$$

$$= MT^{-2}$$

Spring Const

$$F = kx$$

$$\frac{F}{x} = k$$

$$\frac{MKT^{-2}}{L} = k$$

$$k = \underline{MT^{-2}}$$

QUESTIONS [SUMMER SERIES-C- 2025]

The number of significant figure in 0.2370 is:

1

1

2

3

3

4

4

5

QUESTIONS [SUMMER SERIES-X- 2024]

The number of significant numbers in 23.023 is:

1 2

2 3

3 4

4 5 ✓

23.023

Sf = 5

QUESTIONS [SUMMER SERIES-X- 2024]

Planck's constant has same dimensions as that of:

- 1 Energy
- 2 Momentum
- 3 Frequency
- 4 Angular momentum

$$\begin{aligned}
 h &= ML^2T^{-1} \\
 L &= \hbar P \\
 &= L' ML'T^{-1} \\
 &= \underline{ML^2T^{-1}}
 \end{aligned}
 \qquad
 \begin{aligned}
 P &= m\underline{v} \\
 &= \underline{MLT^{-1}}
 \end{aligned}$$

QUESTIONS [SUMMER SERIES-X- 2024]

A force F is given by $F = at + bt^2$, where (t) is time. What are dimensions of a and b ?

1 MLT^{-3} and ML^2T^{-4}

2 MLT^{-3} and MLT^{-4}

3 MLT^{-1} and MLT^0

4 MLT^{-4} and MLT^{-1}

$$F = at + bt^2$$

$$F = at$$

$$MLT^{-2} = aT$$

$$\frac{MLT^{-2}}{T^1} = a$$

$$MLT^{-2}T^{-1} = a$$

$$MLT^{-3} = a$$

$$F = bt^2$$

$$MLT^{-2} = bT^2$$

$$\frac{MLT^{-2}}{T^2} = b$$

$$MLT^{-2}T^{-2} = b$$

$$a = ?$$

$$b = ?$$

$$MLT^{-2-2} = b$$

$$MLT^{-4} = b$$

QUESTIONS [SUMMER SERIES-X- 2024]



In Van der Waal's equation:

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT$$

Where P, V, R, T are pressure, volume, universal gas constant and temperature. What are dimensions of a and b ?

$$P + \frac{a}{V^2}$$

$$P = \frac{a}{V^2}$$

$$\underline{P} \underline{V^2} = a$$

$$M L^{-1} T^{-2} (L^3)^2 = a$$

$$\bullet P = \frac{F}{A} = \frac{MLT^{-2}}{L^2} = ML^{-1}T^{-2}$$

$$V = L^3$$

$$\underline{M L^{-1} T^{-2}} \underline{L^6} = a$$

$$M L^{-1+6} T^{-2} = a$$

$$M L^{-1+6} T^{-2} = a$$

$$M L^{+5} T^{-2} = a$$

$$(V - b)$$

$$V = b$$

$$\underline{L^3} = b$$

QUESTIONS[SUMMER SERIES-Y- 2024]

The number of significant numbers in 0.0003 or 3×10^{-4} is:

1

1

2

2

3

3

4

4

QUESTIONS [SUMMER SERIES-Y- 2024]



Which of the following does not have the dimensions of force?

$$\checkmark F = M L T^{-2}$$

$$\text{Potential gradient} = \frac{V}{l}$$

$$= \frac{M L^2 T^{-3} A^{-1}}{L}$$

$$= M L T^{-3} A^{-1}$$

$$V = \frac{W}{Q}$$

$$I = \frac{Q}{t}$$

$$V = \frac{W}{\underline{I t}}$$

$$\boxed{I t = Q}$$

$$V = \frac{M L^2 T^{-2}}{A T} \quad Q = I t$$

$$V = M L^2 \cancel{T^{-2}} A^{-1} \cancel{T^{-1}} = M L^2 T^{-3} A^{-1}$$

1

Potential gradient

2

Energy gradient

3

Weight

4

Rate of change in linear momentum

QUESTIONS [SUMMER SERIES-Y- 2024]

AIPMT — 2013-14



Q05 A system has basic dimensions as density (d), velocity (v)² and area (A)¹.
Then dimensional representation of force in this system is:

$$F \rightarrow d v A$$

?

1

$$(\underline{A^1 v^2 d^1})$$

2

$$(A^2 v d)$$

3

$$(A v d^2)$$

4

$$(A^2 v^2 d)$$

Sol let

$$F \rightarrow d^a v^b A^c \quad \text{--- (1)} \quad d = \frac{M}{V} = \frac{M}{L^3} = ML^{-3}$$

$$MLT^{-2} \rightarrow (ML^{-3})^a (LT^{-1})^b (L^2)^c$$

$$MLT^{-2} \rightarrow M^a L^{-3a} L^b T^{-b} L^{2c}$$

$$M^1 L^1 T^{-2} \rightarrow M^a L^{-3a} L^b L^{2c} T^{-b}$$

$$\underline{M^1} \underline{L^1} \underline{T^{-2}} \rightarrow \underline{M^a} L^{\underline{-3a+b+2c}} T^{\underline{-b}}$$

compare powers

$$a=1 \quad +b = +2$$

$$\underline{\underline{b=2}}$$

$$v \rightarrow LT^{-1} \quad A \rightarrow L^2$$

$$-3\underline{a}^1 + \underline{b}^2 + 2\underline{c}^1 = 1$$

$$-3 \times 1 + 2 + 2c = 1$$

$$-3 + 2 + 2c = 1$$

$$-1 + 2c = 1$$

$$2c = 1 + 1$$

$$2c = 2$$

$$c = \frac{2}{2} = 1$$

$$a=1$$

$$b=2$$

$$c=1$$

Put value of $a=1$ in equation in Ist
 $b=2$
 $c=1$

$$F \rightarrow \underline{d^a} V^b A^c$$

$$F \rightarrow d'^1 V^2 \underline{A^1}$$

$$F \rightarrow d V^2 A$$

$$F \propto d' V^2 A$$

QUESTIONS [SUMMER SERIES-Y- 2024]

The rotational kinetic energy of a body is given by $\frac{1}{2} I \omega^2$. Use this equation to obtain the dimensions of I . → Neet

• $K.E = \frac{1}{2} I \omega^2$ } $\frac{ML^2 T^{-2}}{T^{-2}} = I$ → Chapter - (8)

$\frac{K.E}{\omega^2} = I$

$\frac{ML^2 T^{-2}}{(T^{-1})^2} = I$

$ML^2 = I$

↓
moment of inertia

$\omega = \frac{\theta}{t}$

$\omega = \frac{1}{T}$

$\omega = T^{-1}$

QUESTIONS[SUMMER SERIES-Z- 2024]

The number of significant numbers in 2.1×10^{-3} is:

1 2 ✓

2 3

3 4

4 5

QUESTIONS [SUMMER SERIES-Z- 2024]

Out of the following pair, which one does not have identical dimensions?

1 Angular momentum and Planck's constant

ML^2T^{-1} ML^2T^{-1}

2 Impulse and momentum

NLM

3 Momentum of inertia and torque

$I = ML^2$ $\tau = ML^2T^{-2}$

4 Work and torque

QUESTIONS [SUMMER SERIES-Z- 2024]

In the relation $y = 2A \sin \omega t \cdot \cos kx$, dimensions of $\frac{y}{A}$ are different than that of:

Chapter (14)

1

ω/k

$$\# \theta = \frac{\text{Arc}}{\text{Radius}} = \frac{k}{x} = 1$$

$$\sin(B)$$

$$\cos(\theta)$$

2

ωt

$$= \sin(\theta) = \frac{P}{H} = \frac{k}{x} = 1$$

$$\sin(A)$$

$$\cos(A)$$

3

$\omega t/kx$

$$= \cos(\theta) = \frac{B}{H} = \frac{k}{x} = 1$$

$$\sin(\omega t)$$

$$\cos(\omega t)$$

4

kx

$$\sin(kx)$$

$$\cos kx$$

$$\sin(A) \quad A \rightarrow 1$$

$$\sin(\omega t) \quad \omega t = 1$$

$$\cos(A) \quad A \rightarrow 1$$

$$\cos(\phi x) \quad \phi x = 1$$

QUESTIONS [SUMMER SERIES-Z- 2024]



The value of Stefan's constant is $a = 5.67 \times 10^{-8} \text{ Js}^{-1}\text{K}^{-4} \text{ m}^{-2}$. find its values in CGS system.

$a = 5.67 \times 10^{-8} \text{ J s}^{-1} \text{ K}^{-4} \text{ m}^{-2} \text{ (SI)}$

$a = 5.67 \times 10^{-8} \text{ J s}^{-1} \text{ K}^{-4} (\text{1m})^{-2}$ 1 Joule = 10^7 erg

$a = 5.67 \times 10^{-8} \times 10^7 \text{ erg s}^{-1} \text{ K}^{-4} (10^2 \text{ cm})^{-2}$ $1 \text{ m} = 100 \text{ cm}$

$a = 5.67 \times 10^{-8} \times 10^7 \text{ erg s}^{-1} \text{ K}^{-4} 10^{-4} \text{ cm}^{-2}$ $1 \text{ m} = 10^2 \text{ cm}$

$a = 5.67 \times 10^{-8} \times 10^7 \times 10^{-4} \text{ erg s}^{-1} \text{ K}^{-4} \text{ cm}^{-2}$

$$a = 5.67 \times \underline{10^{-8}} \times \underline{10^7} \times \underline{10^{-4}} \text{ erg s}^{-1} \text{ K}^{-4} \text{ cm}^{-2}$$

$$a = 5.67 \times 10^{-8+7-4} \text{ erg s}^{-1} \text{ K}^{-4} \text{ cm}^{-2}$$

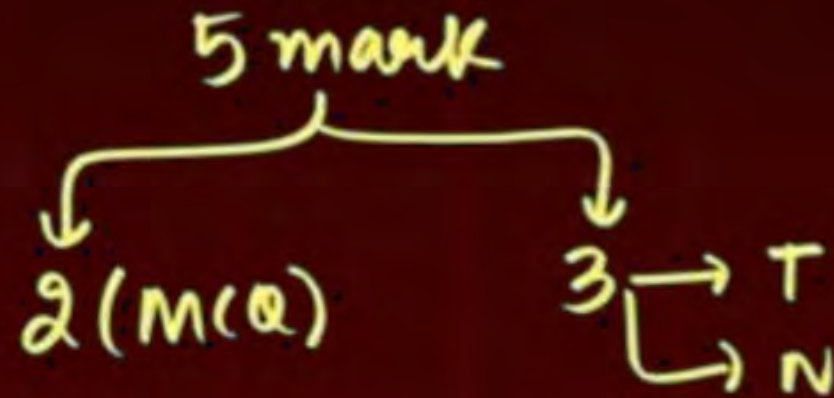
$$a = 5.67 \times 10^{-12+7} \text{ erg s}^{-1} \text{ K}^{-4} \text{ cm}^{-2}$$

$$a = 5.67 \times 10^{-5} \text{ erg s}^{-1} \text{ K}^{-4} \text{ cm}^{-2}$$

PYQ

Numerical

#



Unit I: Physical World and Measurement

05 marks

Units and Measurements

Need for measurement: Units of measurement; systems of units; SI units, fundamental and derived units. Significant figures. Dimensions of physical quantities, dimensional analysis and its applications. *

Roll No.

Z-8-A

Total No. of Questions : 4]

[Total No. of Printed Pages : 8

11th ARF(SZ)JKUT2024-25
208-A

PHYSICS

Time : 3 Hours]

[Maximum Marks : 70

Section-A

(Objective Type Questions)

1 each

1. (i) Which of the following is not the unit of surface tension ? ✓

(A) N/m

(B) J/m²

(C) kg/s²

(D) N/m²

(ii) The numbers 2.745 and 2.735 on rounding off to 3 significant figures

will give :

(A) 2.75 and 2.74

(B) 2.74 and 2.73

(C) 2.75 and 2.73

(D) 2.74 and 2.74

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Z-8-A

208-A

Turn Over

(2)

- (iii) When the distance travelled by a body is proportional to the time taken, what happens to its speed ?
- (A) Becomes zero
 - (B) Increases
 - (C) Remains the same
 - (D) Decreases
- (iv) If a projectile is thrown with velocity u at an angle θ with the horizontal, then velocity at maximum height during projectile motion will be :
- (A) $u \sin \theta$
 - (B) $2u \sin \theta$
 - (C) $2u \cos \theta$
 - (D) $u \cos \theta$
- (v) The forward movement in swimming takes place because of :
- (A) First law of motion
 - (B) Second law of motion
 - (C) Third law of motion
 - (D) Fourth law of motion

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Z-8-A

(vi) The maximum value of static friction when the body is at the verge of starting motion is known as :

- (A) Impending motion
- (B) Angle of repose
- (C) Static friction
- (D) Limiting friction

(vii) Linear velocities of all the particles of the rigid body in rotational motion is :

- (A) One
- (B) Zero
- (C) Same
- (D) Different

(viii) At what depth below the surface of earth the value of ' g ' is same as that of height of 5 km ?

- (A) 10 km
- (B) 7.5 km
- (C) 5 km
- (D) 2.5 km

(4)

(ix) The heat transfer in the light bulb takes place due to :

- (A) Conduction
- (B) Convection
- (C) Radiation
- (D) Both convection and radiation

(x) What is the value of Avogadro's constant ?

- (A) $6.022 \times 10^{23} \text{ mol}^{-1}$
- (B) $6.022 \times 10^{26} \text{ mol}^{-1}$
- (C) $6.022 \times 10^{25} \text{ mol}^{-1}$
- (D) $6.022 \times 10^{22} \text{ mol}^{-1}$

Section-B

(Very Short Answer Type Questions)

2 each

2. (a) From the velocity-time graph of uniform accelerated motion, deduce the equation of motion in velocity and time.
- (b) State the conditions for the equilibrium of a particle under concurrent forces.

11th ARF(SZ)JKUT2024-25 **208-A**
Z-8-A

(5)

- (c) A force of 50 N acts on a body and moves it to a distance of 4 m on a horizontal surface. Calculate work done if the direction of force is at an angle of 60° to the horizontal surface.
- (d) Give two points of difference between conservative and non-conservative forces.
- (e) What is Collision ? Define two types of collision.
- (f) A torque of 20 N-m is applied on a wheel initially at rest. Calculate angular momentum of the wheel after 3 seconds.
- (g) State universal law of gravitation. Derive relation between ' g ' and ' G '.
- (h) State law of equipartition of energy. Also give the concept of mean free path.
- (i) Why longitudinal waves are called pressure waves ?

Section-C

(Short Answer Type Questions)

3 each

3. (a) If string wires of same material of length l and $2l$ vibrate with frequencies 100 Hz and 150 Hz, find the ratio of their tensions.
- (b) Find the dimensions of Planck's constant. If its value in CGS system is 6.62×10^{-27} erg-sec. What will be its value on MKS system ?
- (c) Why are the curved roads banked ? Find the angle of banking of curved road. <https://www.jkboseonline.com>
- (d) State and explain the principle of conservation of angular momentum.
- (e) Define orbital velocity. Derive an expression for it.
- (f) State and explain Hooke's law.
- (g) Define Thermal Equilibrium. How does it occur ? What is the relationship between thermal equilibrium and zeroth law ?

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208-A

Z-8-A

(7)

- (h) State second law of thermodynamics by Kelvin and Clausius. Give two applications of second law of Thermodynamics.
- (i) Show that average kinetic energy of translation per molecule of gas is directly proportional to the absolute temperature of gas.

Section-D

(Long Answer Type Questions)

5 each

4. (a) Show that the path followed by projectile is parabolic when it is projected at an angle θ with horizontal.

Or

Explain cross product of vectors. Give its example and properties.

- (b) State and prove Bernoulli's theorem. Give its applications.

Or

Define Capillarity. Derive ascent formula for the rise of a liquid in a capillary tube and discuss the result obtained.

(8)

(c) What are Standing Waves ? Explain their formation in open organ pipe.

Or

Show that the total energy of particles performing linear simple harmonic motion is constant. What is the time period of total energy in SHM ?

Q/W Find out the SF in given
no.

→ 8834864321

$$SF = 10$$

SUMMARY



Gabber
numerical
session

✓ Tutorial X
Question

$$F = MLT^{-2}$$

$$E = ML^2T^{-2}$$

Thank
You

→ Hand written notes
+
one short
30 min.

♡ gift